

The Hong Kong Polytechnic University

Subject Description Form

Please read the notes at the end of the table carefully before completing the form.

Subject Code	SFT548
Subject Title	Design Creativity in Wearable Technology
Credit Value	3
Level	5
Pre-requisite/ requisite/ Exclusion	N/A
Objectives	Wearable technology is becoming a hot trend. Applications cover various aspects, including medical, healthcare, sportswear, entertaining fashion, smart home textiles and so on. It brings convenience and assists our daily work and thus improves our quality of life. This subject uses design-thinking methodologies to empower students to lead innovation movements using wearable technology. Hence, it aims to enable students to 1) understand how current technology can be used for daily needs and why it is needed from a human-centric point of view, 2) gain hands-on experiences and utilize their creativity on practical design approach integrating textile, design, business and technology to tackle with a real-world problem, and 3) analyze the pros and cons, challenges and opportunities, limitations and future creative developments of wearable technologies, through lectures, tutorials, group discussion, case study and hands-on experience on wearable technology.
Intended Learning Outcomes <i>(Note 1)</i>	Upon completion of the subject, students will be able to: (a) Comprehend the importance and creative trend of wearable technology for a different community of the public, and its technological and cultural meanings to the society (b) Understand the characteristics of various contemporary design methods and technologies in healthcare, medical, sportswear, material science, textile development, sustainability etc. and empathize with users to develop potential solutions towards specific textile issues creatively

	(c) Integrate design thinking methods and technological knowledge to non-linear idea generation and product development processes for commercialization (d) Review the insights gained from current wearable technology products, analyze how design influence technology, and predict future design trend
Subject Synopsis/ Indicative Syllabus <i>(Note 2)</i>	<u>Comprehension of wearable technology knowledge and culture</u> Students will learn about the essential and theoretical principles and the real-world applications of contemporary wearable technology in both literature and physical products. They will study the functions, related trends and aesthetic considerations needed for the market. Moreover, they have to understand its technological and cultural meanings to society. <u>Understanding of the characteristics of contemporary wearable technologies</u> Students will study different wearable technologies and design methods, such as thermal heating, shape memory and lighting, and their development in different aspects. They have to explore multiple factors that affecting the success, including technology development, intellectual protection, aesthetic design, human-centric needs, marketing strategy and so on. Afterwards, they have to utilize the framework of design thinking to discover and define potential technologies to meet current human needs creatively. <u>Integration of design thinking method and technological knowledge</u> Students will study the five stages of design thinking processes, includes, empathize, define, ideate, prototype and test. They will then have a chance to plan and develop a mini wearable technology. Students will be encouraged to have critical thinking and empathy about what is the future trend, what is needed by the market and how it can be realized. They can thus have step-by-step and hands-on experience on advanced design thinking. <u>Review of the insights gained for design planning</u> Students will explore and analyze the strength, weakness, opportunity and threat of wearable technology design, and how design and technology influence each other. They will also discuss issues of data privacy, security, safety and overdependence on technology. Such explorations facilitate discussion on aesthetic, sustainability and ethical consideration, and thus enable students to learn effective future product planning and design.
Teaching/Learning Methodology <i>(Note 3)</i>	Teaching will be comprised of interactive tutorials and lectures for critical analysis and discussions on wearable technology and related design and business issues identified. Students have to analyze different wearable technologies and their potential applications from various points of view. Then, reflect and relate the insights gained to real-world situations, as well as their creative developing

work. They have to prepare and submit a short paper regarding a case study and their reflection. They also have to work in groups for actual design innovation and present their work to enable more case learning, discussion and critical thinking among students.

Assessment Methods in Alignment with Intended Learning Outcomes

(Note 4)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
		a	b	c	d		
1. Group Project Assignment	50%	✓	✓	✓	✓		
2. Individual Assignment	30%	✓	✓	✓	✓		
3. Pre-class preparation, attendance & participation	20%	✓	✓	✓	✓		
Total	100 %						

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

For the group assignment and presentation, students will conduct a mini-design project regarding wearable technology. They then have to submit and present for 15-20 minutes that includes a review of the development background and its significance, design and analysis of the methodology and outcomes, suggestions for commercialization and reflection on how to apply to their future work. Presentation enables more case learning, discussion and critical thinking among students.

For individual assignments, students have to develop an extended piece of writing (1500-2000 words) regarding a chosen case in wearable technology design. This will help to develop and improve the student's level of literacy as well as higher-order critical and creating thinking, advanced literacy and comprehension and abstraction abilities.

For attendance and participation, students will be required to participate in classroom discussions and exercises allowing them to reflect on the research methods, and to discuss the theoretical and technical implications of the design thinking research method.

The grade is calculated according to the percentage assigned; and the completion and submission of all component assignments are required for passing the subject.

The materials submitted for this assessment must be the student's own work. The submitted work may not be accepted for the purpose of assessment if its

	authenticity is questionable. Submitting GenAI-generated materials as students' own work or part of their work is an act of academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions.	
Student Study Effort Expected	Class contact:	
	▪ Lectures	25 Hrs.
	▪ Tutorials	12 Hrs.
	Other student study effort:	
	▪ Individual Assignment	24 Hrs.
	▪ Group Assignment	24 Hrs.
	▪ Self-study & Class Preparation	20 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	Butterfield, Andrew J., and John Szymanski. "Wearable Technology." <i>A Dictionary of Electronics and Electrical Engineering</i> (2018): A Dictionary of Electronics and Electrical Engineering. Web. Tong, Raymond. <i>Wearable Technology in Medicine and Health Care</i>. London, United Kingdom: Academic, an Imprint of Elsevier, 2018. Web. Johnson, Cory., and Bloomberg News. <i>What's next for Wearable Technology?</i> New York: Bloomberg, 2013. Business Education in Video. Web. Holland, Janet. <i>Wearable Technology and Mobile Innovations for Next-generation Education</i>. Hershey, PA: Information Science Reference, 2016. Advances in Educational Technologies and Instructional Design. Web. Information Resources Management Association, Editor. <i>Wearable Technologies: Concepts, Methodologies, Tools, and Applications</i>. Hershey PA: Engineering Science Reference, 2018. Web. Koncar, Vladan. <i>Smart Textiles and Their Applications</i>. Duxford: Woodhead, 2016. Woodhead Publishing in Textiles; No. 178. Web. Schneegass, Amft, Schneegass, Stefan, and Amft, Oliver. <i>Smart Textiles: Fundamentals, Design, and Interaction</i>. Cham: Springer, 2017. Human-computer Interaction Ser. Web. Paret, Dominique, and Pierre Crégo. <i>Wearables, Smart Textiles & Smart Apparel</i>. London, UK: ISTE: Elsevier, 2019. Web. Chapman, R. A. <i>Smart Textiles for Protection</i>. Cambridge, UK: Woodhead	

	Limited, 2013. Woodhead Publishing in Textiles; No. 133. Web. Kirstein, Tünde., and Textile Institute. <i>Multidisciplinary Know-how for Smart Textiles Developers</i>. Oxford; Philadelphia: Woodhead Pub., 2013. Woodhead Publishing in Textiles; No. 139. Web. Kettley, Sarah. <i>Designing with Smart Textiles</i>. London: Fairchild, an Imprint of Bloomsbury, Plc, 2016. Print. Required Reading Range. Course Reader. Tao, and Tao, Xiaoming. <i>Handbook of Smart Textiles</i>. Singapore: Springer, 2016. Springer Reference. Web. Luchs, Swan, Griffin, Luchs, Michael, Swan, Scott, Griffin, Abbie, & Product Development & Management Association. (2016). <i>Design thinking: New product development essentials from the PDMA</i>. Hoboken, New Jersey: John Wiley & Sons. Plattner, H., Meinel, C., & Leifer, L. (2018). <i>Design thinking research: Making distinctions: Collaboration versus cooperation</i> (Understanding innovation). Cham, Switzerland: Springer. Ambrose, G., & Harris, P. (2010). <i>Design thinking</i> (Basics design; 8). Lausanne; London: AVA Academia. Darbellay, F., Lubart, T., & Moody, Z. (2018). <i>Creativity, Design Thinking and Interdisciplinarity</i>. Singapore: Springer Science and Business Media: Springer Nature. Fashioning Tech http://www.fashioningtech.com/ Open Materials http://openmaterials.org/ Sparkfun Electronics http://www.sparkfun.com/
--	--

Note 1: Intended Learning Outcomes

Intended learning outcomes should state what students should be able to do or attain upon subject completion. Subject outcomes are expected to contribute to the attainment of the overall programme outcomes.

Note 2: Subject Synopsis/Indicative Syllabus

The syllabus should adequately address the intended learning outcomes. At the same time, overcrowding of the syllabus should be avoided.

Note 3: Teaching/Learning Methodology

This section should include a brief description of the teaching and learning methods to be employed to facilitate learning, and a justification of how the methods are aligned with the intended learning outcomes of the subject.

Note 4: Assessment Method

This section should include the assessment method(s) to be used and its relative weighting, and indicate which of the subject intended learning outcomes that each method is intended to assess. It should also provide a brief explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes.